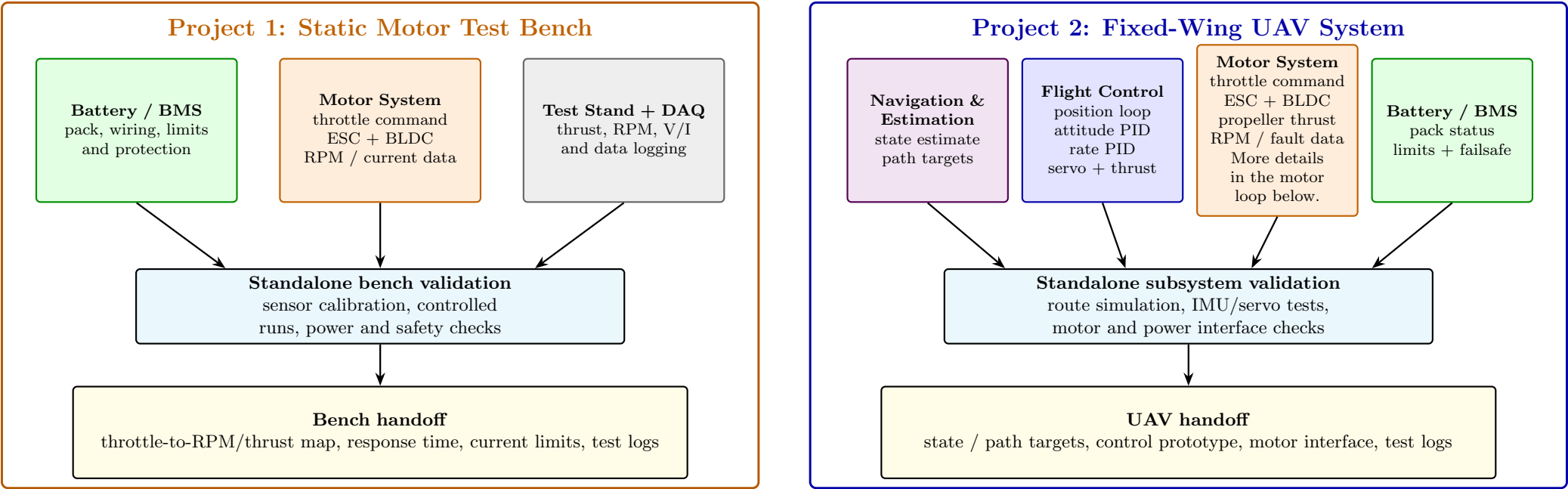
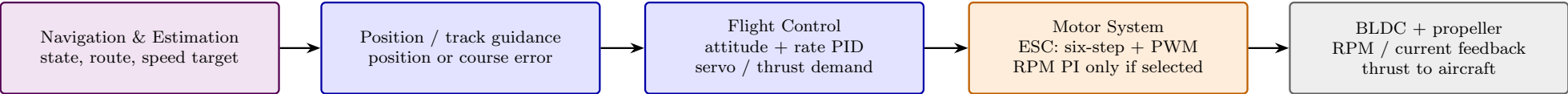


DARES Electronics: Two Projects Before Integration

The work packages below run in parallel. Each project validates its own subsystems and records a handoff package.



Control hierarchy and motor scope



Motor System: which loops to implement?

Function	Decision	Condition / measurement
Six-step + PWM	No: commutation / switching, not a current PI loop	A separate current regulator could be added, but is not shown here.
Speed PI	No: an optional closed-loop RPM controller	Add only when RPM is measurable and holding target RPM is a requirement.
Current limit / faults	No: threshold protection, not a current PI loop	Measure current and voltage; define trip limits and safe response.
FOC current PI	Yes: two inner PI loops for i_d and i_q	Future FOC drive requires phase-current sensing and rotor electrical-angle estimation.
Shaft position loop	Not needed for a continuously spinning propeller	Only add if the motor must hold a specified shaft angle.

Aircraft position / track guidance is outside the Motor System. Label RPM and current as telemetry only where those measurements are actually available.